

Question	Marking guidance	Mark	Comments
04.1	Into plane of the paper ✓	1	
04.2	KE or speed or velocity increases/proton accelerates ✓ (Because) work is done on proton by electric field OR electric field exerts a force on proton (in direction of travel) ✓	2	Accept a description of proton being attracted/repelled by a dee
04.3	Speed/KE constant OR only direction changes OR velocity changes OR moves in (semi-) circle ✓ (Because) magnetic field produces a force (due to motion of proton) OR <u>magnetic</u> force acts (on proton) ✓ Force/acceleration acts at 90° to path/velocity OR reference to Fleming's Left Hand rule OR centripetal force/acceleration acts (on proton) ✓	3	MP1 is for describing motion moving from B to C. MP2 is for explaining origin of force. MP3 is for describing magnetic force as a centripetal force. Accept arguments supported by relevant formulae. Don't give credit for formulae alone. Penalise contradictions within marking points.
04.4	Use of $v = \omega r$ OR $E_k = \frac{1}{2}mv^2$ ✓ Use of both formulae PLUS correct manipulation ✓	2	

Question	Marking guidance	Mark	Comments
04.5	Converts 10 MeV to 1.6×10^{-12} (J) ✓ Substitution into $E_k = \frac{(BqR)^2}{2m}$ or B made the subject $\sqrt{\frac{2mE_k}{q^2R^2}}$ or $\frac{\sqrt{2mE_k}}{qR}$ ✓ 1.4 (T) ✓	3	Condone POT error. Condone 10 (MeV) for E_k. Condone 1.675×10^{-27} or 1.661×10^{-27} for m. Allow 1.3 (T).

Question	Marking guidance	Mark	Comments
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	25	A
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	26	A
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	28	C
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	29	A
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	30	D

6	B
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	27	B
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	28	B
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	29	A
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	30	A
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	32	C
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	vertically downwards/towards S ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	into the page ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	electric force and magnetic force equated and substituted ✓	accept $qE = qvB$ seen (and rearranged) ✓ Do not credit if the charge symbols q are missing	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.4	Use of $E = \frac{V}{d}$ ✓ use of $v = \frac{E}{B}$ to give $v = 9.8 \times 10^3 \text{ (m s}^{-1}\text{)}$ ✓	condone POT error for MP1	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.5	net force/deflection is downwards/towards S ✓ the electric force (upward) is unchanged OR does not depend on speed ✓ the magnetic force (downward) has increased OR magnetic force is proportional to speed ✓	three marking points are independent if MP2 and MP3 are not awarded, one mark can be given for 'magnetic force is greater than electric force'	3	AO3
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	(Reference to Fleming's left-hand rule) Identification that current in slide is towards X and magnetic field is up ✓	Current direction can be drawn on Figure 5	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	(Evidence of use of $F = BIl$) to give $4.8 \times 10^6 \text{ A}$ ✓	Must show some working by either rearrangement or substitution At least 2 sf	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Evidence of use of work done = force $\times$ distance ✓ Evidence of use of GPE gained = mgh ✓ KE = work done – GPE gained ✓ Use of their KE = $\frac{1}{2}mv^2$ to determine v ✓	Expect to see $4.5 (\times 10^9 \text{ J})$ Expect to see $1.0(4) (\times 10^5 \text{ J})$ Do not award MP3 if answer suggests work done = KE gained with no ref to GPE e.g. GPE is negligible Expect to see $8.5 \times 10^3 (\text{m s}^{-1})$	4	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	(doubling rail length) doubles work done on slide and rocket ✓ increasing speed by a factor of (approximately) $\sqrt{2}$ ✓	Condone 'doubles the KE of slide' Allow answer in terms of their value from 03.3	2	1 × AO3 1 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
03.5	Doubling current also doubles B ✓ Force on/work done on of slide and rocket increases by a factor of 4 therefore speed is increased by a factor of (approximately) 2 ✓	Allow in terms of their value from 03.3 Condone 'KE increases by factor of 4' in MP2	2	1 × AO3 1 × AO4
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
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07	D	$\text{kg A}^{-1} \text{s}^{-2}$	AO1
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08	C	0.84 A	AO1
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09	A	$\frac{BQ_v}{m}$	AO1
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16	C	They increase if the peak current in the secondary coil increases.	AO1
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18	B	290	AO2
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Question	Marking guidance	Mark	Comments
01.1	Multiply the number of (vertical) squares by the volts/div ✓ <i>And any two of:</i> Turn off the timebase ✓ Increase the y-gain/decrease the volts/div so the trace takes up more of the screen vertically owtte ✓ Read off the <u>peak-to-peak</u> voltage and halve it ✓	3	Condone 'adjust' for increase/decrease if meaning is clear

Question	Marking guidance	Mark	Comments
01.2	no. of squares found for multiple cycles (accept 9.6 to 10 squares for four cycles here) ✓ their reading of cycle squares multiplied by the timebase (can be implied) OR use of $\omega = 2\pi / T$ ✓ leading to $\omega = 12.7\text{-}13.1 \text{ (rad s}^{-1}\text{)}$ ✓	3	MP1 requires multiple cycles Alternative for MP1, MP2: frequency found using multiple cycles ✓ use of $\omega = 2\pi f$ ✓ At least 3sf

Question	Marking guidance	Mark	Comments
01.3	50 ✓ ms/div(ision) ✓ idea of one cycle filling the screen ✓	3	Expect to see 0.49/10 or 0.049 (s) Accept: an integer multiple of 50 (ms) for mp1 IF idea of having multiple cycles and taking an average given for mp3 ✓✓ MP2 dependent on MP1 Accept ms/cm or ms or s for MP2, if unit matches number

Question	Marking guidance	Mark	Comments
01.4	Use of $\varepsilon_{\max} = BAN\omega$ ✓ $B = 0.039$ (T) ✓ ecf on their value of ω from 1.2	2	Condone rms ε for MP1 0.038 (T) if 13 rad s^{-1} used

Question	Marking guidance	Mark	Comments
01.5	Use of $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$ ✓ Use of $P = \frac{V^2}{R}$ ✓ $P_{\text{mean}} = 1.0(4) \times 10^{-4}$ (W) ✓	3	Full credit for correct answer without working

Total		14
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Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
06.1	Component of its velocity perpendicular to field ✓	1	Accept 'not moving parallel to' magnetic field Reject idea that motion must be perpendicular to the field

Question	Marking guidance	Mark	Comments
06.2	Equates expressions for centripetal force and magnetic force ✓ $p = mv$ substituted and convincing rearrangement ✓	2	Eg $mv^2/r = Bqv$

Question	Marking guidance	Mark	Comments
06.3	Particle loses some of its kinetic energy (to the gas molecules as it travels) ✓ Particle's velocity/momentum decreases ✓ Radius is directly proportional to the velocity/momentum ✓	3	For MP3, accept radius decreases as momentum decreases if supported with equation

Question	Marking guidance	Mark	Comments
06.4	Evidence of determination of a value of the initial radius by direct measurement ✓ Use of $r = mv/Bq$ (condone POT) ✓ Determines mv in range $1.0 \times 10^{-21} \text{ (kg m s}^{-1}\text{)}$ to $1.2 \times 10^{-21} \text{ (kg m s}^{-1}\text{)}$ ✓	3	Evidence must be seen on Figure 11 Give 2 marks max for diameter used instead of radius 2sf max for MP3 This tolerance is adjusted for non-standard printed scripts

Question	Marking guidance	Mark	Comments
06.5	Up to two correct statements and one correct corresponding reason ✓✓✓	3	Gamma photon is neutral or not very ionising (in this medium) because there is no track the charges of the two particles produced are opposite because the spirals are in opposite directions the two particles produced have same initial momentum as the tracks have similar radii (it must be clear which particles are being referred to, which can be implied by describing the track)
Total		12	

Question	Key	Answer
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17	A	N m A^{-1}
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18	C	down the page	into the page
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20	D	0
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	(the particle) moves parallel to the field (line) ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Uses perpendicular velocity ($4.5 \times 10^5 \text{ ms}^{-1}$) AND uses $F = BQv$ ✓ (to get F) Uses their $F = mv^2/r$ ✓ To get $r = 3.9 \times 10^4 \text{ m}$ ✓	Expect to see $8.6 \times 10^{-21} \text{ N}$ Accept use of mv/BQ as evidence of use of equation in MP1 and MP2 Allow ecf for v in MP2	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	(Helix because:) Parallel component of velocity unaffected (by magnetic field) OR (Magnetic) force on particle is perpendicular to its (instantaneous) velocity ✓ So the force is centripetal /the force causes circular motion ✓	Alternative: Clockwise because: Proton motion is in direction of conventional current ✓ Reference to Fleming's LHR ✓	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Refers to emf and rate of change of flux linkage. ✓ High emf achieved by: Rapid (rate of) change (of flux density) owtte ✓ Very large (linked) area as power cables are very long ✓		3	AO3
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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17	A	0.036 N down	AO2
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18	c	0.033 Wb	AO2
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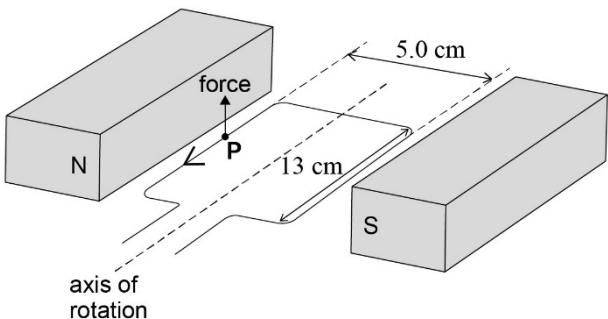
Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Correctly resolves horizontally or vertically ✓ Successfully resolves horizontally to give $\frac{mv^2}{r} = R \sin \theta$ AND vertically to give $mg = R \cos \theta$ ✓ Divides $\frac{mv^2}{r} = R \sin \theta$ by $mg = R \cos \theta$ and rearranges convincingly ✓	Allow full marks from a correct force diagram approach. If no other mark awarded MP1 for $mg \tan \theta = mv^2 \div r$	3	1 AO1 1 AO2 1 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	(Uses $u = \sqrt{rg \tan \theta}$ to find) the velocity of Q = 58(.1) (m s⁻¹) ✓ Uses $t = \frac{s}{v}$ and $s = \pi r$ to find one correct time OR $\omega = \frac{v}{r}$ to find one correct angular speed ✓ All correct plus reasoned conclusion that Q passes B first ✓	P: 20(.1) (s) Q: 19.6 (s) $\omega_P = 0.156 \text{ (rad s}^{-1}\text{)}$ $\omega_Q = 0.160 \text{ (rad s}^{-1}\text{)}$ Expect to see a reference to the different times in MP3	3	1 AO1 1 AO2 1 AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Clear attempt to take moments ✓ Recognises that $R_2 = mg$ OR $F_2 = \frac{mv^2}{r}$ when speed = V ✓ Uses both $R_2 = mg$ and $F_2 = \frac{mv^2}{r}$ plus clear manipulation ✓	Each term must be a force x a distance expect to see $R_2 \frac{w}{2} = F_2 h$ but accept additional moments terms for MP1	3	1 AO2 2 AO3
Total			9	
Question	Answers	Additional comments/Guidelines	Mark	AO

13	D	The forces on the horizontal sides of the coil are in opposite directions.	AO3

16	C	It decreases to a constant negative value.	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Current drawn as an arrow pointing along left-hand wire towards the front ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Any two from: ✓✓ • use of $F = BIL$ (with either distance) • use of couple = Fs (with the other distance) • use of $\tau = rF$ couple = 0.044 (N m) ✓		3	AO2

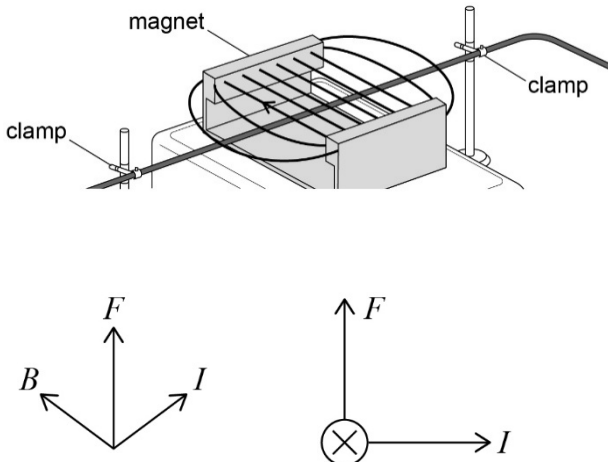
Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	$\varepsilon = Blv$ or $75 (\times 10^{-3}) \times 8.75 (\times 10^{-2}) \times 2.3$ seen ✓ 0.015 (V) ✓		2	1 × AO1 1 × AO2
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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16	B	The kinetic energy of the particle is constant.	AO1
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17	C	
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Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	(The magnetic flux density) that produces a force of 1 N on a 1 m wire with a current of 1 A going through it ✓ ...with the idea that the magnetic field and current are perpendicular to each other ✓	Allow per 1 m and per 1 A All of 1 N , 1 A and 1m are needed for mp1 Alternative answer for mp1 using $F = Bqv$ (The magnetic flux density) that produces a force of 1 N on a 1 C charge travelling at a speed of 1 m s⁻¹ through the field ✓ For this alternative, mp2 can be 'velocity' in place of 'current' OWTTE	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Arrow drawn on Figure 1 on a magnetic field line pointing towards the back pole AND an explanation which relates to Fleming's left-hand rule ✓ Force on the wire stated as being up (or shown in a labelled diagram) AND a reference made to Newton's third law ✓ Direction of current stated clearly and correctly in explanation (or shown in a diagram) ✓	 mp2 can be awarded for a clearly labelled force acting upwards on the wire in Figure 1 or in a diagram in the answer showing the force acting upwards on the wire or on a labelled diagram for Fleming's left-hand rule. mp3 can be awarded for current direction shown on Figure 1. If more than one arrow is provided, it must be clear which is the magnetic field (and which is the current)	3	1 × AO1 1 × AO2 1 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Use of $F = B I l$ equated to mg ✓ 0.044 (T) ✓	Allow POT error on mp1 $B = \frac{mg}{Il} = \frac{0.45 \times 10^{-3} \times 9.81}{2.0 \times 5.0 \times 10^{-2}} = 0.0441 \text{ T}$	2	1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	The first wire has an additional force on it from the extended wire outside of the poles (and this gives an incorrect value of B when calculated) ✓ The idea that the vertical sections of the wire will not affect the reading on the balance / not have a vertical force on them ✓	For mp1 need to see an explanation of the additional force on the first wire Do not allow statements that the vertical sections of the wire are outside of the field (so there is no force on them)	2	AO4

Total			9	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Correct readings from graph: 0.38 to 0.40 T at 0.05 s and -0.38 to -0.40 T at 0.15 s OR $\Delta B = 0.76$ to 0.80 T ✓ Use of $\varepsilon = \frac{\Delta BAN}{\Delta t}$ ✓ $\varepsilon_1 = 0$ (V) and $\varepsilon_2 = 0.96$ (V) ✓	$\varepsilon = \frac{\Delta BAN}{\Delta t} = \frac{0.80 \times 1.5 \times 10^{-4} \times 800}{0.10} = 0.96 \text{ V}$ Allow ε_2 consistent with their graph readings in range 0.91 to 0.96 V Ignore minus sign in final answers	3	1 × AO2 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	The idea that the amplitude is reduced because the (maximum) change in magnetic flux happens over a longer time ✓ The idea that the amplitude is reduced because there is a smaller (maximum) change in magnetic flux ✓ The frequency of the emf will reduce because $f = \frac{1}{T}$ ✓	Allow answers that use the $BAN\omega$ equation for mp1 and mp2 as long as the explanation is clear Allow references to peak emf in place of amplitude Allow “frequency and period are inversely proportional” for mp3 as an alternative to citing the equation	3	1 × AO2 2 × AO3

Total			6	
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Question	Key	Answer	AO
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18	B	$\frac{2\pi m}{BQ}$	AO2
		proton	AO3

19	B		
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Unknown Unknown - Unit 3 - Q04

question	marking guidance	mark	comments
04.1	1 N per A per m or 1 Wb m⁻² or quotes: $B = F/IL$ with terms defined or induced $EMF = \Delta BAN/t$ with terms defined or a slightly flawed attempt at the definition in statement form ✓ (1 only) It is the flux density (perpendicular to a wire) that produces a force of 1 N per m on the wire when the current is 1 A ✓✓ (both marks) or $B = F/IL$ and 1 T is flux density when $F = 1N$; $I = 1A$ and $L = 1 m$ or induced $EMF = \Delta BAN /t$ and 1 T is the flux change when emf = 1V for $A=1$ $N=1$ and $t=1$ or similar	2	
04.2	force on charge due to E field , $F_E = Eq$ or Vq/d and force due to B field, $F_B = Bqv$ or $Eq=Bqv$ ✓ cancels q and states explicitly $v = \frac{E}{B}$ or $\frac{V}{dB}$ ✓	2	
04.3	$v = 20000/0.14$ (seen) or 143×10^3 (m s⁻¹) ✓	1	

Unknown Unknown - Unit 3 - Q04 (continued 2)

04.4	$Bqv = mv^2/r$ or $r = mv/Bq$ (allow e instead of q) ✓ mass of ion = $1.7 \times 10^{-27} \times 58$ (may be in equation) or $(9.86 \times 10^{-26} \text{ kg seen})$ or radius = 0.14 m (may be in equation) ✓ Substitutes and arrives at 0.62 to 0.63 (T) ✓	3	
04.5	Calculates new radius (0.145 m) or diameter (0.288 m) using $r \propto m$ or otherwise allowing ecf ✓ 0.010 m (condone 0.01 m) or 0.0096 – 0.0097 m (Allow 0.0079 m or 0.008 m due to use of different sfs for B and v) ✓	2	

question	marking guidance	mark	comments
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13	A
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